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ParaDigMa: a Python toolbox for extracting Parkinson disease digital biomarkers from daily life wrist sensor data

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Summary

ParaDigMa is an open-source Python package designed to extract digital biomarkers of Parkinson disease (PD) from wrist-worn sensor data collected during passive monitoring in daily life (Post, Veldkamp, et al., 2025). It processes raw accelerometer, gyroscope and photoplethysmography (PPG) signals using modular pipelines which address common challenges in real-world wearable sensor data analysis. The package currently targets three clinically relevant symptom domains: (1) tremor, (2) arm swing during gait, and (3) pulse rate (to derive digital biomarkers related to autonomic dysfunction). ParaDigMa enables researchers to derive interpretable and clinically meaningful digital biomarkers to monitor PD in daily life.

Statement of need

The use of wearable sensors to monitor PD in daily life has grown exponentially over the past two decades (Deb et al., 2022), with mounting evidence supporting their utility for diagnosis, monitoring disease progression, and evaluating treatment efficacy (Sun et al., 2024). However, the extraction of digital biomarkers from raw sensor signals often relies on proprietary or non-replicable pipelines applied to individual cohorts. To support their adoption in clinical and research practice, and to enable meaningful comparisons across studies, reproducible and standardized data processing workflows are essential (Boyle et al., 2025; Sun et al., 2024).

ParaDigMa addresses this need by providing pipelines for extracting digital biomarkers of PD from wrist-worn sensor data obtained from passive monitoring in daily life. Instead of requiring researchers to build end-to-end pipelines from scratch, ParaDigMa streamlines the transformation of raw wrist sensor data into clinically meaningful outcomes. In addition, its modular and standardized architecture enables users to replace or extend components, encouraging collaboration to expand and refine the toolbox for broader use in digital health research.

Comparison

While several toolboxes exist for analyzing wearable sensor data from individuals with PD, ParaDigMa distinguishes itself through its focus on passive monitoring in daily life using wrist-worn devices.

Two comparable toolboxes are Mobgap (Küderle et al., 2025) and GaitPy (Czech & Patel, 2019), tailored to gait analysis using lower-back sensors. In contrast, ParaDigMa extracts digital biomarkers from wrist-worn sensors, enabling analysis of domains not adequately captured by lower-back sensors, such as tremor and reduced arm swing. An additional advantage of tailoring to wrist-worn devices is the high availability and unobtrusiveness of smartwatches, facilitating long-term monitoring in free-living conditions. Another key differentiator is ParaDigMa's potential for cross-domain symptom analysis, crucial for the heterogeneous symptomatic manifestation and progression of PD (Maetzler et al., 2009). This enables development of composite digital biomarkers—a step toward personalized progression monitoring (Adams et al., 2023). A PD-specific toolbox covering multiple symptomatic domains is PDKit (Stamate et al., 2021), including modules for gait and tremor. It focuses on extracting features from sensor data obtained from active monitoring (i.e., the analysis of brief, scripted tasks), where the execution of the activities is highly standardized. In contrast, ParaDigMa is tailored to passive, continuous monitoring in the person's own home environment (were the precise nature of the actually performed activities is unknown) and includes functionalities to handle noisy, unscripted real-world sensor data. Furthermore, ParaDigMa requires one wrist-worn sensor to extract digital biomarkers of a variety of symptomatic domains, while PDKit uses different sensor placements per domain.

Beyond PD-specific toolboxes , ParaDigMa also differs from general-purpose toolboxes for processing wearable sensor data. For instance, HeartPy (van Gent et al., 2019) enables users to derive insights related to pulse rate, but lacks PD-specific preprocessing, such as handling PPG motion artifacts in PD (for example introduced by tremor (Veldkamp et al., 2025)). Another example is GGIR (Migueles et al., 2019), developed in R, which extracts generic activity metrics from accelerometer data collected during passive monitoring. However, this toolbox similarly does not contain PD-specific functionalities. A final example is mHealth (Snyder et al., 2020), developed in R, which contains sensor position-agnostic functionalities but is limited to preprocessing and feature extraction.

Architecture

ParaDigMa is a Python toolbox for processing and analyzing wrist-worn sensor data. Its architecture, illustrated in Figure 1, is designed for flexibility and ease of use: each processing step is implemented as a standalone function, enabling users to run complete pipelines through a small number of sequential function calls. The toolbox supports data from inertial measurement units (IMUs) and/or photoplethysmography (PPG) sensors, and is device-agnostic, accepting input with minimal requirements (specified in its documentation).

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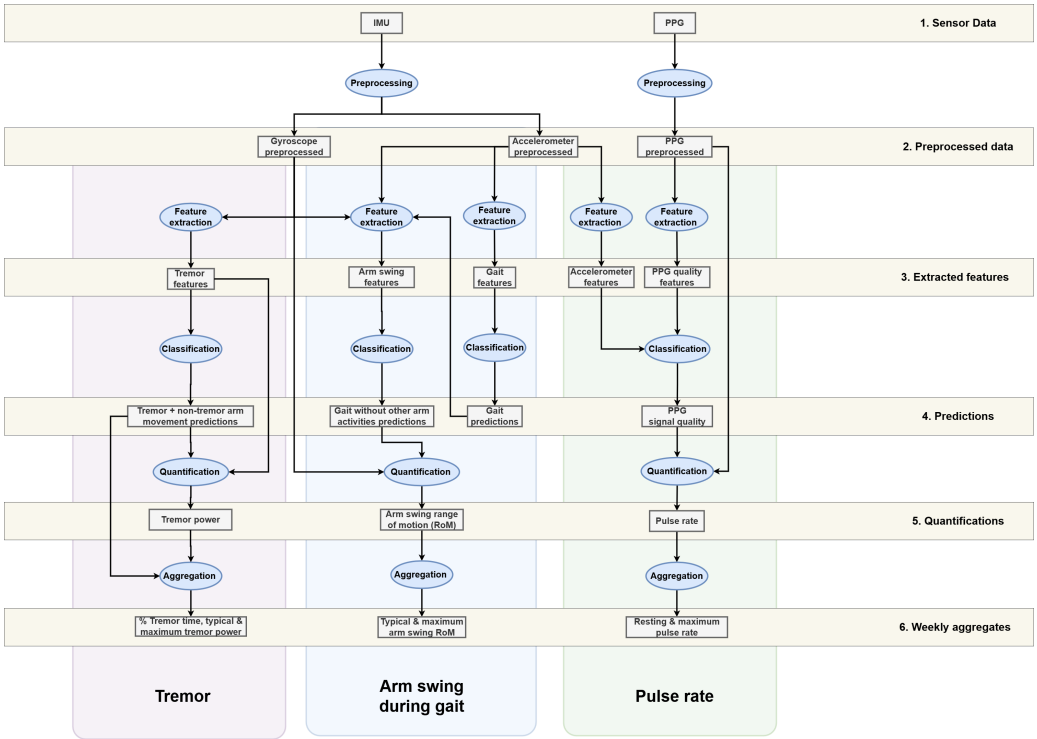

Figure 1: A schematic overview of the architecture of ParaDigMa. Horizontal bars indicate categories of input or output data, with rectangles representing specific data objects. Data processing steps, shown as blue ovals, follow a consistent structure across domains. Arrows represent the sequential flow of operations, and colored columns distinguish the individual processing domains.

From top to bottom, data are processed as follows. First, raw sensor data are preprocessed through resampling, interpolation, and filtering (1. Preprocessing). The resulting time series are segmented into fixed-size windows, from which temporal and spectral features are extracted (2. Feature extraction). These features are then passed to domain-specific classifiers for the detection of relevant behavior (e.g., gait) or symptoms (e.g., tremor) (3. Classification). These predictions are combined with the preprocessed data to quantify the detected behavior (e.g., arm swing range of motion during gait) or symptoms (e.g., tremor power) (4. Quantification). The last step (5. Aggregation) contains functionalities to aggregate the output from steps 3 and 4 across relevant time scales to derive digital biomarkers (e.g., weekly typical tremor power).

This sequential, modular approach enables users to extract the output after each processing step. For example, users can examine the presence of tremor, the quantity of gait with or without other arm activities, and the PPG signal quality after classification (step 3). Furthermore, after quantification (step 4), users can examine daily fluctuations in tremor power (estimated per 4 seconds), arm swing range of motion (estimated per 1.5 seconds), and pulse rate (estimated per 2 seconds). Or, to study disease progression, users can assess week-to-week longitudinal changes in tremor time, typical and maximum tremor power, typical and maximum arm swing range of motion, or resting and maximum pulse rate after aggregation (step 5).

The ParaDigMa pipelines have been developed and validated using representative data from the Parkinson@Home Validation Study (Evers et al., 2020) and the Personalized Parkinson Project (PPP) (Bloem et al., 2019). The cross-sectional validity of all three pipelines has been demonstrated in recent scientific publications (Post, van Laarhoven, et al., 2025; Timmermans et al., 2025; Veldkamp et al., 2025). Ongoing work is focused on longitudinal validation of the pipelines, including assessments of test-retest reliability and sensitivity to disease progression.

To facilitate standardization and reproducibility, ParaDigMa includes optional functionality for working with Time Series Data Format (TSDF) (Kok et al., 2024). TSDF provides a consistent and efficient format for storing and reading time series data along with relevant metadata, such as data source and temporal coverage. This makes it easier to document, cache, and reuse intermediate outputs across pipeline stages, and to maintain interoperability across studies or devices.

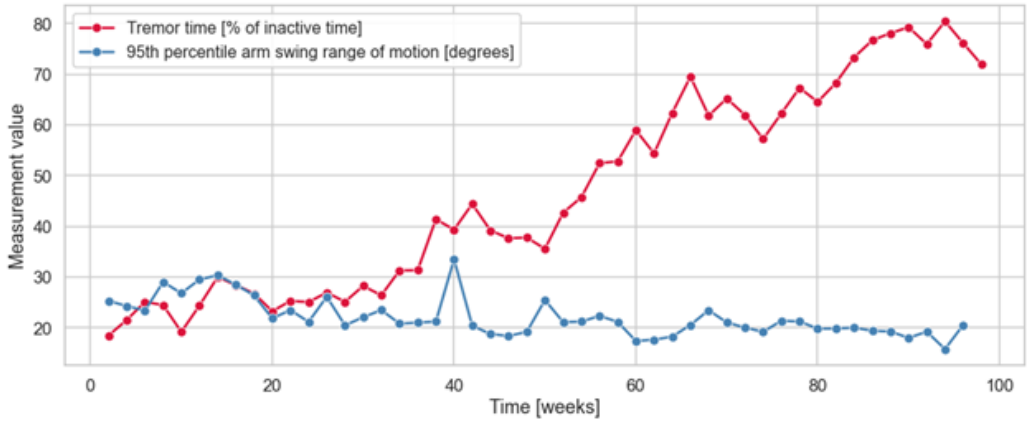

Figure 2: An example of bi-weekly measurements of ParaDigMa from a participant in the Personalized Parkinson Project study. For both the tremor and arm swing during gait pipelines a single digital biomarker is shown.

Application

Driven by the need for more sensitive biomarkers of therapeutical efficacy and disease progression, the use of wearable sensors in the context of clinical trials is rapidly increasing (Masanneck et al., 2023), providing numerous opportunities for the application of ParaDigMa. Currently, ParaDigMa is being used in multiple clinical trials to derive exploratory endpoints, including the Slow-SPEED trial (Radboud University Medical Center, 2025), which evaluates the effects of an exercise intervention, and the Orchestra trial, which investigates the effects of an alpha-synuclein misfolding inhibitor on PD progression (UCB Biopharma SRL, 2024). In addition, ParaDigMa is being applied in large cohort studies to better understand the heterogeneity of PD progression, including the PPP (Bloem et al., 2019) and Parkinson's Progression Markers Initiative (PPMI) (Marek et al., 2011) cohort studies.

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References

Adams, J. L., Kangarloo, T., Tracey, B., O'Donnell, P., Volfson, D., Latzman, R. D., Zach, N., Alexander, R., Bergethon, P., Cosman, J., Anderson, D., Best, A., Severson, J., Kostrzebski, M. A., Auinger, P., Wilmot, P., Pohlson, Y., Waddell, E., Jensen-Roberts, S., … Ray Dorsey, E. (2023). Using a smartwatch and smartphone to assess early Parkinson's disease in the WATCH-PD study. *Npj Parkinson's Disease*, *9*(1), 64. https://doi.org/10.1038/s41531-023-00497-x

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Bloem, B. R., Marks, W. J., Silva De Lima, A. L., Kuijf, M. L., van Laar, T., Jacobs, B. P. F., Verbeek, M. M., Helmich, R. C., Van De Warrenburg, B. P., Evers, L. J. W., intHout, J., van de Zande, T., Snyder, T. M., Kapur, R., & Meinders, M. J. (2019). The Personalized Parkinson Project: Examining disease progression through broad biomarkers in early Parkinson's disease. *BMC Neurology*, *19*(1), 160. https://doi.org/10.1186/s12883-019-1394-3

Boyle, L. D., Giriteka, L., Marty, B., Sandgathe, L., Haugarvoll, K., Steihaug, O. M., Husebo, B. S., & Patrascu, M. (2025). Activity and Behavioral Recognition Using Sensing Technology in Persons with Parkinson's Disease or Dementia: An Umbrella Review of the Literature. *Sensors*, *25*(3), 668. https://doi.org/10.3390/s25030668

Czech, M. D., & Patel, S. (2019). GaitPy: An Open-Source Python Package for Gait Analysis Using an Accelerometer on the Lower Back. *Journal of Open Source Software*, *4*(43), 1778. https://doi.org/10.21105/joss.01778

Deb, R., An, S., Bhat, G., Shill, H., & Ogras, U. Y. (2022). A Systematic Survey of Research Trends in Technology Usage for Parkinson's Disease. *Sensors*, *22*(15), 5491. https://doi.org/10.3390/s22155491

Evers, L. J. W., Raykov, Y. P., Krijthe, J. H., Silva De Lima, A. L., Badawy, R., Claes, K., Heskes, T. M., Little, M. A., Meinders, M. J., & Bloem, B. R. (2020). Real-Life Gait Performance as a Digital Biomarker for Motor Fluctuations: The Parkinson@Home Validation Study. *Journal of Medical Internet Research*, *22*(10), e19068. https://doi.org/10.2196/19068

Kok, P., Kasalica, V., & Rodríguez-Sánchez, P. (2024). *Tsdf*. Zenodo. https://doi.org/10.5281/ZENODO.13745614

Küderle, A., Tasca, P., Bicer, M., Kirk, C., Megaritis, D., Hinchliffe, C., Stihi, A., Muecke, A., Babar, Z., Kluge, F., Mueller, A., Mazzà, C., Del Din, S., Cereatti, A., Rochester, L., Rooks, D., & Caulfield, B. (2025). *MobGap*. Zenodo. https://doi.org/10.5281/ZENODO.15673267

Maetzler, W., Liepelt, I., & Berg, D. (2009). Progression of Parkinson's disease in the clinical phase: Potential markers. *The Lancet Neurology*, *8*(12), 1158–1171. https://doi.org/10.1016/S1474-4422(09)70291-1

Marek, K., Jennings, D., Lasch, S., Siderowf, A., Tanner, C., Simuni, T., Coffey, C., Kieburtz, K., Flagg, E., Chowdhury, S., Poewe, W., Mollenhauer, B., Klinik, P.-E., Sherer, T., Frasier, M., Meunier, C., Rudolph, A., Casaceli, C., Seibyl, J., … Taylor, P. (2011). The Parkinson Progression Marker Initiative (PPMI). *Progress in Neurobiology*, *95*(4), 629–635. https://doi.org/10.1016/j.pneurobio.2011.09.005

Masanneck, L., Gieseler, P., Gordon, W. J., Meuth, S. G., & Stern, A. D. (2023). Evidence from ClinicalTrials.gov on the growth of Digital Health Technologies in neurology trials. *Npj Digital Medicine*, *6*(1), 23. https://doi.org/10.1038/s41746-023-00767-1

Migueles, J. H., Rowlands, A. V., Huber, F., Sabia, S., & Hees, V. T. van. (2019). *GGIR: A Research Community–Driven Open Source R Package for Generating Physical Activity and Sleep Outcomes From Multi-Day Raw Accelerometer Data*. https://doi.org/10.1123/jmpb.2018-0063

Post, E., van Laarhoven, T., Raykov, Y. P., Little, M. A., Nonnekes, J., Heskes, T. M., Bloem, B. R., & Evers, L. J. W. (2025). Quantifying arm swing in Parkinson's disease: A method accounting for arm activities during free-living gait. *Journal of NeuroEngineering and Rehabilitation*, *22*(1), 37. https://doi.org/10.1186/s12984-025-01578-z

Post, E., Veldkamp, K., Timmermans, N., Soriano, D., Kasalica, V., Kok, P., Cagnan, H., Bloem, B. R., Little, M. A., Raykov, J. P., van Laarhoven, T., & Evers, L. J. W. (2025). *ParaDigMa: A toolbox for deriving Parkinson's disease Digital Markers from real-life wrist sensor data*. Zenodo. https://doi.org/10.5281/ZENODO.15223364

DRAFT

Radboud University Medical Center. (2025). *Slow-SPEED-NL: Slowing Parkinson's Early Through Exercise Dosage-Netherlands* (Clinical Trial Registration No. NCT06193252). clinicaltrials.gov.

Snyder, P., Tummalacherla, M., Perumal, T., & Omberg, L. (2020). *Mhealthtools: A Modular R Package for Extracting Features from Mobile and Wearable Sensor Data*. Zenodo. https://doi.org/10.5281/ZENODO.3731707

Stamate, C., Pons, J. S., Weston, D., & Roussos, G. (2021). PDKit: A data science toolkit for the digital assessment of Parkinson's Disease. *PLOS Computational Biology*, *17*(3), e1008833. https://doi.org/10.1371/journal.pcbi.1008833

Sun, Y., Wang, Z., Liang, Y., Hao, C., & Shi, C. (2024). Digital biomarkers for precision diagnosis and monitoring in Parkinson's disease. *Npj Digital Medicine*, *7*(1), 218. https://doi.org/10.1038/s41746-024-01217-2

Timmermans, N. A., Terranova, R., Soriano, D. C., Cagnan, H., Raykov, Y. P., Bucur, I. G., Bloem, B. R., Helmich, R. C., & Evers, L. J. W. (2025). A generalizable and open-source algorithm for real-life monitoring of tremor in Parkinson's disease. *Npj Parkinson's Disease*, *11*(1), 205. https://doi.org/10.1038/s41531-025-01056-2

UCB Biopharma SRL. (2024). *A Double-Blind, Placebo-Controlled, Randomized, 18-Month Phase 2a Study to Evaluate the Efficacy, Safety, Tolerability, and Pharmacokinetics of Oral UCB0599 in Study Participants With Early Parkinson's Disease* (Clinical Trial Registration No. NCT04658186). clinicaltrials.gov.

van Gent, P., Farah, H., van Nes, N., & van Arem, B. (2019). HeartPy: A novel heart rate algorithm for the analysis of noisy signals. *Transportation Research Part F: Traffic Psychology and Behaviour*, *66*, 368–378. https://doi.org/10.1016/j.trf.2019.09.015

Veldkamp, K. I., Evers, L. J. W., Laarhoven, T. van, Raykov, Y. P., Little, M. A., Ho, K. C., Shin, S., Lehmann, L. S., Bloem, B. R., Brouwer, M. A., & Thannhauser, J. (2025). *Heart rate monitoring using wrist photoplethysmography in Parkinson disease: Feasibility and relation with autonomic dysfunction* (p. 2025.08.15.25333751). medRxiv. https://doi.org/10.1101/2025.08.15.25333751

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